

A Study on Scalable Parallelism and Integrity in Honeycomb Networks

– Sustaining Pedestrian/Bike Friendly Urbanism

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Abstract

Due the concern on environment issues, establishing pedestrian/bike friendly urbanism is widely encouraged. For busy traffic areas, regular degree-3 DSRC (dedicated short range communication) networks are suggested to support availability, reliability, and maintainability in paths of multi-lanes. Applying alternative network topologies, such as spider-webs, generalized honeycomb tori, for comparing, analyzing are proposed in DSRC and cellular communications to enhance reliability, integrity in communications. In this paper, we find developing scalable parallelism or mutually independent Hamiltonian paths in Honeycomb torus is possible. It is aimed to promote safety-assured, mobile communication environments, efficient, reliable maintenance and information integrity, especially in highly possible interfered places.

Keywords: *dedicated short range communication (DSRC); Integrity; maintainability; parallelism, interference*

1 Introduction

In the last decade, communication systems have been advancing at startlingly rates with faster, more reliable and economical subsystems. The wireless communication infrastructure can help customers get plenty of information for flexibly managing tasks and enjoying amenities. To promote customer satisfaction ubiquitously, coordinating bipartite networks, i.e., the spider-web (SW) network, the generalized honeycomb torus (GHT) network, and the hypercube-derived network (CCC, cube-connected cycles), are proposed for establishing integrity assured platforms in dynamical, challenging environments.

Except for occlusion and material interference; however, intentional radio intrusions such as those caused by truck drivers' personal privacy devices (PPD) are getting more attention [1, 15]. Facing

issues like energy conservation, establishing pedestrian/bike friendly urbanism can be a global mission. Hence, to well support easy, orderly, safety-quality-assured traffic and (bicycle/car) parking spaces, with possibilities of mobile business services along paths, wireless communication integrity should be seriously concerned.

To obtain better telecommunication integrity, using correlated data analysis is rational [4, 17]. Hence, communication transceivers or sensors allocated, networked in complementary positions are basically prototyped, probably in modular perspective- just like coordinated sensory mechanisms in two eyes', ears' promoting more diagnosing resilience, intelligence. Furthermore, such dual-surveillance can incorporate with allocation design on concerning availability and maintainability in changing environments.

An effective, efficient sequential order for area inspection needs to be pondered especially if available resources are limited [13]. On networking, a Hamiltonian path (cycle) is a path (cycle) linking all nodes once. Therefore, such sequentially ordered feature is essential in communication due to confidential needs in network inspection, maintenance errands. Wireless communications may have peer-to-peer connectivity. Hence, different mathematical performance can be assumed, inspected to help assure network reliability, and analyze the impact of communication interference confidentially.

On the strategy of tracing performance on different networks with same wireless communication nodes, we show that the performance of mutually independent Hamiltonian paths (MIHP) in both GHT and SW [6, Fig.1, 2, 3] networks can be feasible approaches. Their topology can be adaptable in the dedicated short range communication (DSRC) and the cellular communication networks. Network topologies have comprehensive information capability can be considered first.

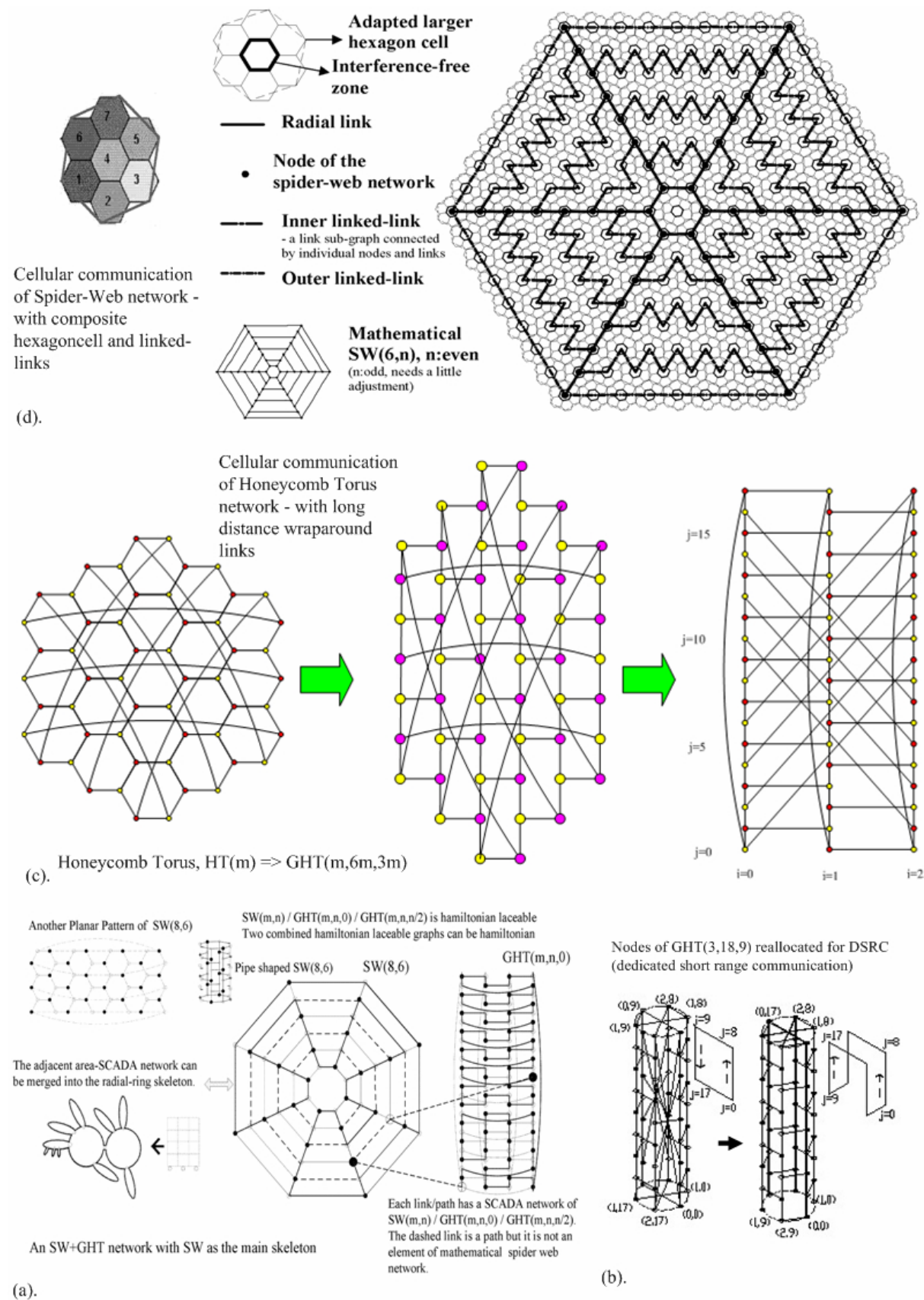


Fig. 1. Link adaptability in wide area communication network, (a). urban dedicated short range communication network with Spider-Web networks' integration and fractal connectivity performances, (b). wireless communication quality affected by link adaptability, (c). cellular communication modelled by honeycomb tori, GHT networks, (d). cellular communication modelled by Spider-Web networks.

2 Mathematical Preliminaries

Communication networks are usually illustrated by graphs, in which nodes represent processors and edges represent links between processors. Let $G=(V,E)$ be a graph if V is a finite set and E is a subset of $\{(a,b) \mid (a,b) \text{ is an unordered pair of } V\}$. A path is delimited by $(x_0, x_1, x_2, \dots, x_{n-1})$. A path is called a Hamiltonian path if its nodes are distinct and span V . A cycle is a path of at least three nodes such that the first node is the same as the last node. A cycle is called a Hamiltonian cycle or Hamiltonian if its nodes are distinct except for the first node and the last node, and if they span V [5].

A bipartite graph $G = (V, E)$ is a graph such that $V = A \cup B$ and E is a subset of $\{(a, b) \mid a \in A \text{ and } b \in B\}$; if $G - F$ remains Hamiltonian for any $F = \{a, b\}$ with $a \in A$ and $b \in B$, then G is l_p -Hamiltonian. A graph G is l -edge Hamiltonian if $G - e$ is Hamiltonian for any $e \in E$; moreover, if there is a Hamiltonian path between any pair of nodes $\{c, d\}$ with $c \in A$ and $d \in B$, then the bipartite graph G is Hamiltonian laceable.

The number of links connecting a node is called the degree; networks regularly have fewer degrees and are generally economic [18]. For DSRC's functionality in busy traffic conditions, regular degree-3 is minimal for supporting availability, reliability, and maintainability in paths of multi-lanes; consequently, networks prototyped for DSRC are regular degree-3 based in this paper.

Assume that m and n are positive integers, where n is even and $m \geq 2$. Let d be any integer such that $(m-d)$ is even. The *generalized honeycomb torus* [2], $GHT(m, n, d)$ is the graph with the node set $\{(i, j) \mid 0 \leq i < m, 0 \leq j < n\}$ such that (i, j) and (k, l) are adjacent if they satisfy one of the following conditions: (1) $i=k$ and $j=l \pm 1 \pmod{n}$; (2) $j=l$ and $k=i-1$ if $i+j$ is even; and (3) $i=0, k=m-1$, and $l=j+d \pmod{n}$ if j is even.

The spider-web network, $SW(m, n)$ is a graph with the node set $\{(i, j) \mid 0 \leq i < m, 0 \leq j < n\}$, where $m, n \geq 4$ even integers, such that (i, j) and (k, l) are adjacent if they satisfy one of the following conditions: (1) $i=k$ and $j=l \pm 1$; (2) $j=l$ and $k=i+1 \pmod{m}$ if $i+j$ is odd or $j=n-1$; and (3) $j=l, k=i-1 \pmod{m}$ if $i+j$ is even or $j=0$. The configuration of $SW(m, n)$ is shown in Fig. 1 (a, d).

It is proved that $GHT(m, n, 0)$, also named $HReT(m, n)$, is l -edge Hamiltonian if $n \geq 4$; l_p -Hamiltonian if $n \geq 6$ or $m=2, n \geq 4$ [3] When m and n are positive integers with $n, m-n/2$ being even, $GHT(m, n, n/2)$ is proved l -edge Hamiltonian if $n \geq 4$; l_p -Hamiltonian if $n \geq 6$ or $m=2, n \geq 4$ [7] When $m, n \geq 4$ even integers, $SW(m, n)$ is proved l -edge Hamiltonian; l_p -Hamiltonian [10]. Thus, the fault-tolerance in which we are engaged is

systematically based. Moreover, $GHT(m, n, 0)$, $GHT(m, n, n/2)$, and $SW(m, n)$ are Hamiltonian laceable if $m, n \geq 4$ integers [9, 11]

Two Hamiltonian paths $P_1=(u_1, u_2, \dots, u_{n(G)})$ and $P_2=(v_1, v_2, \dots, v_{n(G)})$ of G from u to v are independent if $u=u_1=v_1, v=u_{n(G)}=v_{n(G)}$, and $u_i \neq v_i$, for every $1 < i < n(G)$. A set of Hamiltonian paths, $\{P_1, P_2, \dots, P_k\}$, of G from u to v is *mutually independent* if any two different Hamiltonian paths are independent from u to v . The mechanism of *mutually independent Hamiltonian paths (MIHP)* can be applied to parallel processing [19]. Such a feature is also considered for secret communications [8, 14]. It is proved that $SW(m, n)$ has MIHP performance [6]; and it is believed that $GHT(m, n, 0)$ and $GHT(m, n, n/2)$ can have such performance if nodes are enough (e.g. more than 16).

3 Network Proposal - sustaining pedestrian/bike friendly urbanism

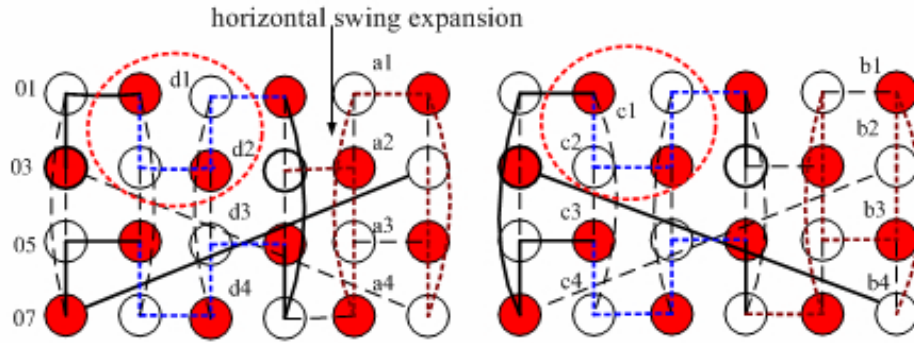
Recently, due the concern on energy conservation, global warmth, establishing pedestrian, bike friendly urbanism, tourism is encouraged by a lot of countries. Two directions are concerned. One is on occlusion and interference issues; the other is on maintainability.

From small settlements to big cities, the traditional cities generally have relatively convenient accessibilities, and enough living resources, and then it can be developed step by step - inherently in a radial-ring configuration, probably including water ways, centered in the original settlement [12]. By inherent, real radial-ring motion flow, a grid urban pattern can enlarge its original scope of urban activities. In this perspective, a grid pattern can rationally be adapted to a radial-ring pattern; moreover, the grid can be illustrated to be the radial-ring [12]. It is found that bipartite spider-web networks have mathematical laceability (Fig. 1 (a)) to sustain the ring performance of connected spider-web networks.

The proposed radial-ring configured DSRC network prototype is integrated via the spider-web network; in other words, it can be a spider-web (radial-ring) network of spider-web (radial-ring) networks. Furthermore, just as cities may be multi-centered, flexible amount of radial-rings of radial rings can be composed or sub-divided (e.g., in a campus site, a large vehicle architecture); i.e., fractal. Such connected rings can have sequential order, fault-tolerance for maintenance, and fractal features for hierarchical management.

Exemplary concepts for establishing MIHP:

1. Special node; 2. Indivisibility; 3. Consistency.



concept of indivisibility

03, 01, 02, 08, 06, 05, 07, 04 Lt.

03, 08, 06, 05, 07, 01, 02, 04 Rt.

special node set
(scalable) - example

consistency (direction and order, adaptability may be allowed):

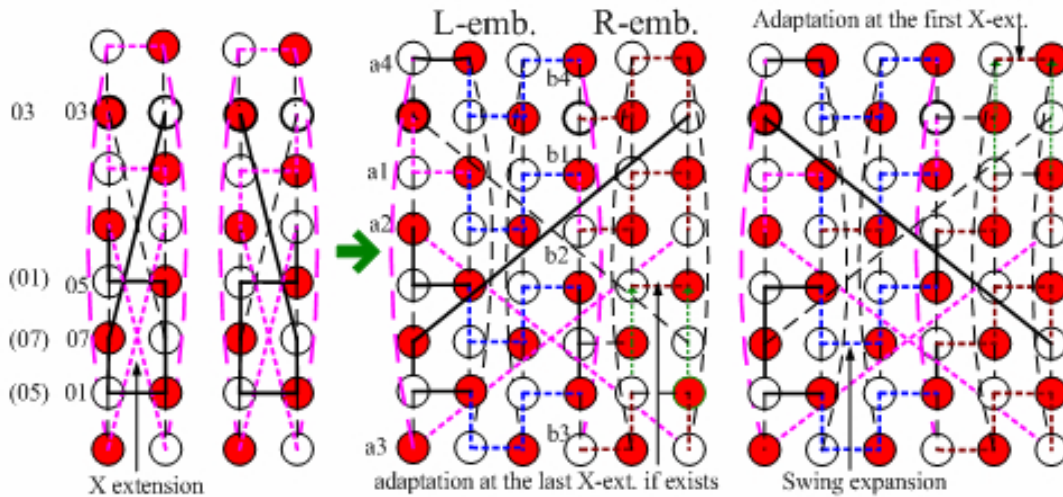
03, 01, 02, 08, 06, 05, 07, b2, b3, b4, b1, a1, a4, a3, a2, 04 Lt.

03, b4, b1, b2, b3, a3, a2, a1, a4, 08, 06, 05, 07, 01, 02, 04 Rt.

(a). without vertical extension (X-ext.)

Exemplary concepts for establishing MIHP:

1. X-extension; 2. Balance; 3. Adaptation.



Basic pattern 03, 05, 06, 08, 02, 01, 07, 04 Lt.

(proved) 03, 08, 02, 01, 07, 05, 06, 04 Rt.

with X-ext. 03, a1, b1, b2, a3, a4, b4, b3, a2, 05, 06, 08, 02, 01, 07, 04 Lt.

(proving) 03, 08, 02, 01, 07, 05, 06, b2, a3, a4, b4, b3, a2, a1, b1, 04 Rt.

(b). with vertical extension (X-ext.)

Fig.2 Proof of two MIHP in GHT ($m, n, n/2$), m even, $n \geq 4$, $x_1 - x_2$ odd, $y_1 = y_2$ even.

In another perspective, historical heritage and other cultural, natural assets are often resources for tourism development. However, such assets may affect flows of spatial development. Except for adopting underground or overpass construction technologies, such assets can be more economically concerned with, or preserved if the proposed information network prototype can be applied to prevent adversary effects on traffic flows and spatial development - i.e., proposed information networks can be well connected either underground or overpass to help traffic well moved.

Parallelism along paths is aimed to analyze spatial-tempo interference. Moreover, the system manager may use such parallelism for operating dynamic authentication/authorization in transporting privacy demanded tasks, which can be recorded through active RFID (radio frequency identification) systems. This paper proposes a network in which the sequence of such orderly operation can be both flexibly and logically adjusted in a Hamiltonian way, because the network has scalability and Hamiltonian laceability.

Hence, the aforementioned, highly reliable security-information networks can offer continuous and thoughtful protection for travelers. That is vital in planning future urban infrastructure. Such infrastructure can offer another benefit, specifically on bikes' parking, which can be organized in parking spaces, definitely not on the sidewalks. Moreover, enhancing more flexibility in infrastructure can be considered via networking; parking structures may provide an alternative place for freight distribution and economical synergies [16].

3-2. Linking alternatives

Proposed network prototypes can have different mathematical performances. Hence, a set of nodes can be linked differently to form or reside in different network topologies. Then, such performances can be compared, analyzed to enhance communication integrity, reliability. For example:

- (1). GHT ($m,n,0$) and GHT($m,n,n/2$) are different, but in the radio transmission in DSRC, their communication restrictions can be similar (Fig. 1 (a)).
- (2). On cellular communication, both SW($6,n$) and GHT($m,6m,3m$) are proposed (Fig. 1(c,d)) with MIHP, fault-tolerance performances.
- (3). Radial-ring, SW networks, have more fault-tolerance than ring networks. The area DSRC network can be configured as an integral SW network. Naturally, radial-ring can be

grouped, and even configured as a radial-ring of radial-rings (Fig. 1(a)).

4 Proof Examples (two MIHP on GHT($m,n,n/2$))

GHT($m,n,n/2$) can have scalable MIHP performance if the quantity of nodes is large enough. In this paper, an approach is introduced. We can establish basic two MIHP first, and then check their vertical extension and horizontal expansion. Assume (x_1,y_1) and (x_2,y_2) are two end nodes of the basic pattern, and they are based on the basic definition of GHT($m,n,n/2$). ($0,0$) is the left-lowest node's coordinate, but an alternative node number sequence is used to show no collision in two Hamilton path patterns.

A. m even, $n=4$; $y_1=y_2$, even; (x_2-x_1) odd

Although GHT($2,4,2$) has previously been assigned a (x,y) coordinate presentation, i.e., coordinates ($0,0$) and ($1,3$) being arranged at the lowest-left and highest-right positions, respectively, another orderly presentation is shown as Fig. 2 to illustrate two Hamiltonian paths (as Lt. and Rt. patterns) are mutually independent from end nodes, 03 [i.e., coordinate ($0,2$)] and 04 [i.e., coordinate ($1,2$)].

The horizontal embeddings, L-embeddings and R-embeddings are studied to show the scalability performance of mutually independent Hamiltonian paths (MIHP) existed in ($m, n, n/2$), m even, for end nodes with a same, even y coordinate. First, consider $n=4$ as Fig.2 (a), the underline shows the location of L-embeddings and R-embeddings, and the italic underline is for the R-embedding. For the L-embedding, the node contents, including series of the typical node sequence, within 01,02 and 06,05 of both patters are "exactly" same; hence, 01,02 or 06,05 can be deemed as a special node set or two special nodes, e.g., 06 and 05. Apparently, 01,02 can keep lagging in the Lt. pattern. The sequential difference, 2, of the 08, 07, or 06,05 within both patterns can not be equaled by adding nodes of the multiples of four. For the R-embedding, one embedding is shown as the below; its consistency property can help collision free as the below (Fig. 2(a)):

03,01,02,08,06,05,07,b2,b3,b4,b1,a1,a4,a3,a2,04 Lt.
03,b4,b1,b2,b3,a3,a2,a1,a4,08,06,05,07,01,02,04 Rt.

Then, by X shaped vertical extensions (namely, X-extensions), the proven MIHP performance in basic patterns of GHT($m,4,2$) can be morphologically extended to GHT(m,n,d), $n>4$, $d=n/2$, $n \bmod(4)=0$ (or $n-d$ even) (Fig. 2(b)); i.e., the linking relationship associated with $d=n/2$ can be naturally maintained along with the inserting to

increase n . The degree-3 mechanism can make the extension sequence fixed in repeated “vertical, non-vertical” rhythm. Hence, both patterns’ X-ext. are generally consistent. By analyzing balancing of adding nodes, development sequential B. m even, $n \geq 4$; $y_1 = y_2$, odd; $(x_2 - x_1)$ odd

The basic Lt. and Rt. Hamiltonian path patterns in a GHT(2,12,6) network can have no collision except for end nodes as Fig. 3.

Both X-extensions, noted as the subscript x , are in consistent directions. Node blocks are deemed as special nodes. On the Lt. nodes 07, 09,10, 08, 19, 17,18, 20, 22,21 keep lagging, and nodes 23, 12, 14, 16, 03, 01,02, 24, 11, 13, 15, 04 keep leading. Nodes

directions, and some adaptations, we can find that MIHP performance exists in GHT($m,n,n/2$), m even, $n \geq 4$, $x_1 - x_2$ odd, $y_1 = y_2$ even.

within 15,04 (24,11) (23,12) of the Lt. keep leading of their compared nodes within 04,15 (11,24) (12,23) of the Rt.

Assume $R\text{-emb.} \geq L\text{-emb.}$ first. Nodes within 08,19 of the Lt. keep lagging of their compared nodes within 04,15 of the Rt. Nodes within 20,07 of the Rt. Then, assume $R\text{-emb.} < L\text{-emb.}$ Nodes within 08,19 of the Lt. keep indivisibility of their compared nodes within 04,15 of the Rt. As for nodes within X-ext. similar as the example B, they can keep collision free.

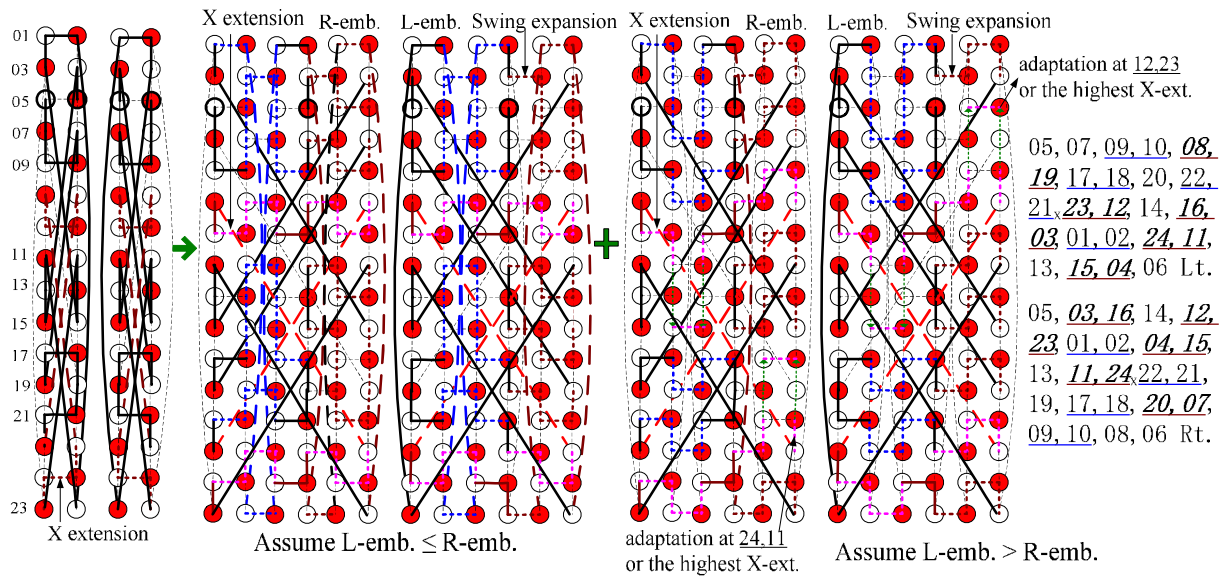


Fig. 3. MIHP performance may exist in HT(2) / GHT(2,12,6).

5 Conclusions

To promote comfort, safety-assured, mobile communication environments for even pedestrians and cyclists, efficient, reliable maintenance and information integrity in communication networks need be concerned, especially in highly possible interfered places, including parking garages. For DSRC in busy traffic conditions, regular degree-3 networks with Hamiltonian performance are suggested for supporting availability, reliability, and maintainability in paths of multi-lanes. SW networks having more fault-tolerance than ring networks are suggested as a mechanism to integrate paths’ DSRC networks. SW(6, n) and GHT($m,6m,3m$) with MIHP, fault-tolerance performances are proposed to enhance reliability, integrity in cellular communications. In this paper, we find developing scalable parallelism or mutually independent Hamiltonian paths in

Honeycomb torus is possible.

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